Question	Answer	Marks
1	**Defining the problem**	
	L is the independent variable and *f* is the dependent variable, or vary *L* and measure *f.*	**1**
	Keep ρ constant	**1**
	Methods of data collection	
	Labelled diagram of workable experiment including: • rod supported by string / elastic bands from a clamp • clamp attached to stand, with stand on bench • two labels from stand, clamp, hammer, microphone, rod, string.	**1**
	Diagram showing labelled microphone connected to labelled oscilloscope.	**1**
	Method to measure *L*, e.g. use a metre rule	**1**
	Method to measure mass (*m*) (of metal rod), e.g. use a (top-pan) balance	**1**
	Method of Analysis	
	Plots a graph of log *f* against log *L* or equivalent e.g. log *f* against log $\frac{1}{L}$	**1**
	n = – gradient (for log *f* against log $\frac{1}{L}$: *n* = gradient)	**1**
	$E = 4\rho \times 10^{2\times y\text{-intercept}}$ (for lg *f* vs lg $\frac{1}{L}$: $E = 4\rho \times 10^{2\times y\text{-intercept}}$) (for ln *f* against ln *L*; $E = 4\rho \times \mathrm{e}^{2\times y\text{-intercept}}$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations** Any **six** from:	**6**
	D1 Precaution linked to falling rod, e.g. sand tray / cushion (in case rod falls) OR gently hit rod prevent rod falling	
	D2 Method to determine area of rod (A) e.g. measure diameter (d) of rod using a micrometer / calipers	
	D3 Repeat measurements of diameter along the length of rod / around the rod and average diameter	
	D4 Method to determine ρ from experimental method, e.g. $\rho = \frac{m}{AL}$ **and** $A = \frac{\pi d^2}{4}$ **or** $\rho = \frac{4m}{\pi d^2 L}$ **or** $r = \frac{d}{2}$ **and** $\rho = \frac{m}{\pi r^2 L}$	
	D5 Perform experiment in a quiet room	
	D6 Reasoned method to prevent rod hitting microphone, e.g. have a gap between rod and microphone / gently hit rod **or** method to obtain measurable signal from the microphone, e.g. use a cone to increase the sound detected by the microphone	
	D7 Method to determine frequency from oscilloscope, e.g. T = time-base × (horizontal) length (of one wave) **and** $f = 1/T$	
	D8 Method to reduce uncertainties e.g. use large values of L to reduce (percentage) uncertainty in L **or** adjust time-base to display as few waves as possible or Z waves on oscilloscope and divide time by Z **or** wait for the wave(form) / frequency to stabilise (and reach resonance)	
	D9 Repeat measurements of f for each value of L **and** average f	

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Question	Answer	Marks
1	D10 Relationship is valid <u>if</u> a straight line is produced (passing through $\log\left(\frac{1}{2}\sqrt{\frac{E}{\rho}}\right)$ OR $\frac{1}{2}\log\left(\sqrt{\frac{E}{4\rho}}\right)$). Do not accept passing through the origin.	

Question	Answer	Marks
2(a)	Gradient = $\frac{3}{E}$ *y*-intercept = $\frac{4Z}{E}$	**1**
2(b)	$\frac{1}{I}$ / A^{-1} 4440 or 4444 5410 or 5405 6250 or 6250 7140 or 7143 8000 or 8000 8700 or 8696	**1**
	Uncertainties in $\frac{1}{I}$ From ± 90–110 to ± 360–400	**1**

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